

Optimal Geometric Charts for Anchored Divided-Difference Iterations

Local Linearization, Curvature Cancellation, and Jet-Optimal Preconditioning

Ivan Besevic

Independent Mathematical Research
github.com/ivan-fr/poussiere_de_x

May 2026

Abstract

Anchored Pandrosion for a scalar analytic equation $G(z) = 0$ can be written, in a chart ϕ , as the fixed-anchor divided-difference map

$$P_{\phi,a}(s) = a - \frac{G(\phi(a))}{(G \circ \phi)[a, s]}.$$

The local behavior of the raw fixed-anchor iteration is therefore determined by the transformed equation $F_\phi = G \circ \phi$. This note develops a geometric theory of chart choice. At a simple root ρ with root coordinate $r = \phi^{-1}(\rho)$,

$$P'_{\phi,a}(r) = 1 - \frac{F'_\phi(r)}{F_\phi[a, r]}, \quad P'_{\phi,a}(r) = \frac{F''_\phi(r)}{2F'_\phi(r)}(a - r) + \mathcal{O}((a - r)^2) \quad (a \rightarrow r).$$

Hence chart optimization is equivalent to flattening F_ϕ near the root. We prove a first-order optimality criterion: among root-normalized charts, the linear multiplier becomes $\mathcal{O}((a - r)^2)$ if and only if $F''_\phi(r) = 0$, equivalently

$$G''(\rho)(\phi'(r))^2 + G'(\rho)\phi''(r) = 0.$$

This identifies a curvature-balancing law for the chart. We also show that the local inverse chart of G linearizes the transformed equation exactly and yields one-step convergence, while its quadratic jet gives a canonical practical approximation. Python-generated figures for $G(z) = e^z - 2$ illustrate the difference between affine, curvature-balanced, and exact inverse charts. The main message is conceptual: before iterating, choose coordinates in which the transformed equation is as linear as possible.

1 Introduction

The monomial Pandrosion scheme and its later analytic fixed-anchor formulation show that the essential denominator is a divided difference, not a derivative. In the general scalar analytic setting, one studies

$$P_{F,a}(s) = a - \frac{F(a)}{F[a, s]}, \quad F[a, s] = \frac{F(a) - F(s)}{a - s},$$

with a fixed anchor a satisfying $F(a) \neq 0$. The local residual factor at a simple root is explicit, and in the analytic continuation paper this residual law was transported through coordinate charts. The natural next question is therefore geometric rather than algorithmic:

Which charts make the anchored divided-difference map as contractive as possible?

This question is not cosmetic. In the raw fixed-anchor iteration, the multiplier is linear in the anchor distance. A well-chosen chart can reduce the coefficient of that linear term, or even cancel it. The central idea of the present note is that chart optimization should be read as *local linearization of the transformed equation*. Once the equation is made flat enough in the chosen coordinates, the anchored iteration inherits the improvement automatically.

The paper is local and deliberately modest. It does not claim a global best chart for arbitrary nonlinear equations. Instead, it gives a local geometric criterion, an ideal inverse-chart model, a canonical second-order jet approximation, and a numerical illustration.

Main contributions. Let G be analytic near a simple zero ρ . The results below establish the following.

- (i) The charted anchored multiplier depends only on $F_\phi = G \circ \phi$.
- (ii) Among root-normalized charts, first-order optimality is equivalent to $F_\phi''(r) = 0$.
- (iii) This is equivalent to a curvature-balancing condition

$$G''(\rho)(\phi'(r))^2 + G'(\rho)\phi''(r) = 0.$$

- (iv) The local inverse chart of G gives the ideal model $F_\phi(s) = s - r$, hence one-step anchored convergence.
- (v) The quadratic jet of that inverse is the canonical first nontrivial approximation to the ideal chart.

2 Anchored divided differences in a chart

Let $U \subset \mathbb{C}$ be a domain, let $F : U \rightarrow \mathbb{C}$ be analytic and nonconstant, and let $a \in U$ satisfy $F(a) \neq 0$.

Definition 2.1 (Anchored divided-difference map). The anchored fixed-point map associated with (F, a) is

$$P_{F,a}(s) = a - \frac{F(a)}{F[a, s]}, \quad F[a, s] = \frac{F(a) - F(s)}{a - s}, \quad (1)$$

where $F[a, a] = F'(a)$ is inserted at the removable point.

Now let G be analytic on a domain $\Omega \subset \mathbb{C}$, let $\phi : V \rightarrow \Omega$ be a local biholomorphic chart, and define

$$F_\phi(s) = G(\phi(s)).$$

Fix an anchor $a \in V$ such that $G(\phi(a)) \neq 0$. The charted anchored map is simply the anchored map for F_ϕ :

$$P_{\phi,a}(s) = a - \frac{G(\phi(a))}{F_\phi[a, s]}. \quad (2)$$

Thus all local questions reduce to the transformed equation F_ϕ .

Proposition 2.2 (Exact multiplier and local expansion). *Let F be analytic near a simple zero r , let $F(a) \neq 0$, and let $a \neq r$. Then r is a fixed point of $P_{F,a}$ and*

$$P'_{F,a}(r) = 1 - \frac{F'(r)}{F[a, r]}. \quad (3)$$

Moreover, as $a \rightarrow r$,

$$P'_{F,a}(r) = \frac{F''(r)}{2F'(r)}(a - r) + \mathcal{O}((a - r)^2). \quad (4)$$

Proof. The fixed-point claim and the exact formula are standard for the fixed-anchor secant map: if $F(r) = 0$ then $F[a, r] = F(a)/(a - r)$ and substitution gives $P_{F,a}(r) = r$. Differentiating (1) at r yields (3). For the local expansion, write

$$F(a) = F'(r)(a - r) + \frac{F''(r)}{2}(a - r)^2 + \mathcal{O}((a - r)^3),$$

so that

$$\frac{(a - r)F'(r)}{F(a)} = 1 - \frac{F''(r)}{2F'(r)}(a - r) + \mathcal{O}((a - r)^2).$$

Since $P'_{F,a}(r) = 1 - (a - r)F'(r)/F(a)$, formula (4) follows. $\square$

Applied to $F_\phi = G \circ \phi$, this gives

$$P'_{\phi,a}(r) = \frac{F''_\phi(r)}{2F'_\phi(r)}(a - r) + \mathcal{O}((a - r)^2), \quad r = \phi^{-1}(\rho), \quad G(\rho) = 0. \quad (5)$$

This already identifies the correct optimization target: the chart should make $F''_\phi(r)$ small.

3 Root-normalized charts and first-order optimality

The right class of coordinates is obtained by normalizing the first jet at the root.

Definition 3.1 (Root-normalized chart). Let ρ be a simple zero of G . A local chart ϕ is *root-normalized at ρ* if there exists a coordinate $r \in V$ such that

$$\phi(r) = \rho, \quad F'_\phi(r) = G'(\rho)\phi'(r) = 1.$$

Equivalently, in the chart coordinate the transformed equation has unit first derivative at the root.

For a root-normalized chart, Taylor expansion gives

$$F_\phi(s) = (s - r) + \frac{F''_\phi(r)}{2}(s - r)^2 + \frac{F'''_\phi(r)}{6}(s - r)^3 + \mathcal{O}((s - r)^4). \quad (6)$$

Hence the coefficient $F''_\phi(r)/2$ is the first nonlinear defect of the transformed equation.

Theorem 3.2 (First-order optimality criterion). *Let G be analytic near a simple zero ρ , and let ϕ be a root-normalized chart with root coordinate r . The following are equivalent.*

- (a) $P'_{\phi,a}(r) = \mathcal{O}((a - r)^2)$ as $a \rightarrow r$.
- (b) $F''_\phi(r) = 0$.
- (c) The chart satisfies the curvature-balancing identity

$$G''(\rho)(\phi'(r))^2 + G'(\rho)\phi''(r) = 0. \quad (7)$$

In that case the transformed equation is linear up to third order:

$$F_\phi(s) = (s - r) + \mathcal{O}((s - r)^3).$$

Proof. For a root-normalized chart, (5) becomes

$$P'_{\phi,a}(r) = \frac{F''_{\phi}(r)}{2}(a-r) + \mathcal{O}((a-r)^2).$$

Thus (a) is equivalent to $F''_{\phi}(r) = 0$, which proves (a) $\Leftrightarrow$ (b). Next,

$$F''_{\phi}(r) = G''(\rho)(\phi'(r))^2 + G'(\rho)\phi''(r)$$

by the chain rule. Hence $F''_{\phi}(r) = 0$ is exactly (7), proving (b) $\Leftrightarrow$ (c). The final claim follows from the Taylor expansion (6). $\square$

Remark 3.3 (Geometric meaning). The quantity ϕ''/ϕ' is the logarithmic curvature of the chart. Theorem 3.2 says that the chart should bend so as to cancel the quadratic curvature of the original equation. Affine scaling can change the size of the multiplier, but a genuinely curved chart is needed to cancel the first nonlinear defect unless $G''(\rho) = 0$ already.

Corollary 3.4 (Canonical quadratic jet). *Let G be analytic near a simple zero ρ , and choose the root coordinate $r = 0$. The quadratic jet*

$$\phi_{\text{jet}}(s) = \rho + \frac{s}{G'(\rho)} - \frac{G''(\rho)}{2G'(\rho)^3}s^2 + \mathcal{O}(s^3) \quad (8)$$

is root-normalized and first-order optimal. It is the unique quadratic jet with these two properties.

Proof. The root-normalization imposes $\phi(0) = \rho$ and $\phi'(0) = 1/G'(\rho)$. Theorem 3.2 then forces

$$\phi''(0) = -\frac{G''(\rho)}{G'(\rho)^3}.$$

This uniquely determines the quadratic jet. $\square$

4 The ideal inverse chart

The previous section gives a second-order criterion. There is also an ideal local model.

Theorem 4.1 (Exact linearization by the local inverse chart). *Let G be analytic near a simple zero ρ . Then there exists a local inverse Ψ of G near 0 with $\Psi(0) = \rho$. If one defines, in a root coordinate with $r = 0$,*

$$\phi_*(s) = \Psi(s), \quad (9)$$

then

$$F_{\phi_*}(s) = G(\phi_*(s)) = s$$

exactly on a neighborhood of 0. Consequently, for every anchor a in that neighborhood,

$$P_{\phi_*,a}(s) = 0$$

for all s where the map is defined. In the ideal inverse chart, anchored iteration converges in one step.

Proof. Since $G'(\rho) \neq 0$, the inverse function theorem gives a local inverse Ψ with $G(\Psi(s)) = s$. Hence $F_{\phi_*}(s) = s$. Its divided difference is identically 1, so

$$P_{\phi_*,a}(s) = a - \frac{F_{\phi_*}(a)}{F_{\phi_*}[a, s]} = a - a = 0.$$

$\square$

Remark 4.2 (Interpretation). The exact inverse chart is not usually available in closed form for a general analytic equation. Nevertheless, it provides the correct conceptual target: the best chart is the one that makes the transformed equation as close as possible to the identity map. Corollary 3.4 is precisely the second-order truncation of this ideal model.

A simple one-parameter family already exhibits the optimization mechanism.

Proposition 4.3 (Quadratic family). *Assume $r = 0$ and let*

$$\phi_\eta(s) = \rho + \frac{s}{G'(\rho)} + \eta s^2. \quad (10)$$

Then ϕ_η is root-normalized for every η , and the first-order multiplier coefficient is

$$P'_{\phi_\eta, a}(0) = c_\eta a + \mathcal{O}(a^2), \quad c_\eta = \frac{G''(\rho)}{2G'(\rho)^2} + \eta G'(\rho). \quad (11)$$

The unique first-order optimal parameter is

$$\eta_* = -\frac{G''(\rho)}{2G'(\rho)^3}. \quad (12)$$

Proof. Since $\phi_\eta(0) = \rho$ and $\phi'_\eta(0) = 1/G'(\rho)$, the family is root-normalized. Moreover,

$$\phi''_\eta(0) = 2\eta.$$

Substituting this into (7) yields

$$F''_{\phi_\eta}(0) = \frac{G''(\rho)}{G'(\rho)^2} + 2\eta G'(\rho),$$

so the coefficient in (5) is exactly $c_\eta = F''_{\phi_\eta}(0)/2$. Setting $c_\eta = 0$ gives (12). $\square$

5 Numerical illustration: $G(z) = e^z - 2$

We now illustrate the theory on the simple analytic equation

$$G(z) = e^z - 2, \quad \rho = \log 2.$$

Since $G'(\rho) = G''(\rho) = 2$, the root-normalized affine chart is

$$\phi_{\text{aff}}(s) = \rho + \frac{s}{2},$$

and the first-order optimal quadratic jet from (8) is

$$\phi_{\text{opt}}(s) = \rho + \frac{s}{2} - \frac{s^2}{8}.$$

The ideal inverse chart is explicit here because the inverse of $e^z - 2$ near 0 is

$$\phi_*(s) = \log(2 + s).$$

Therefore

$$\begin{aligned} F_{\text{aff}}(s) &= 2e^{s/2} - 2 = s + \frac{s^2}{4} + \mathcal{O}(s^3), \\ F_{\text{opt}}(s) &= 2e^{s/2 - s^2/8} - 2 = s - \frac{s^3}{24} + \mathcal{O}(s^4), \end{aligned}$$

$$F_*(s) = s.$$

The affine chart leaves a quadratic defect, the optimal quadratic chart cancels it, and the inverse chart removes it exactly.

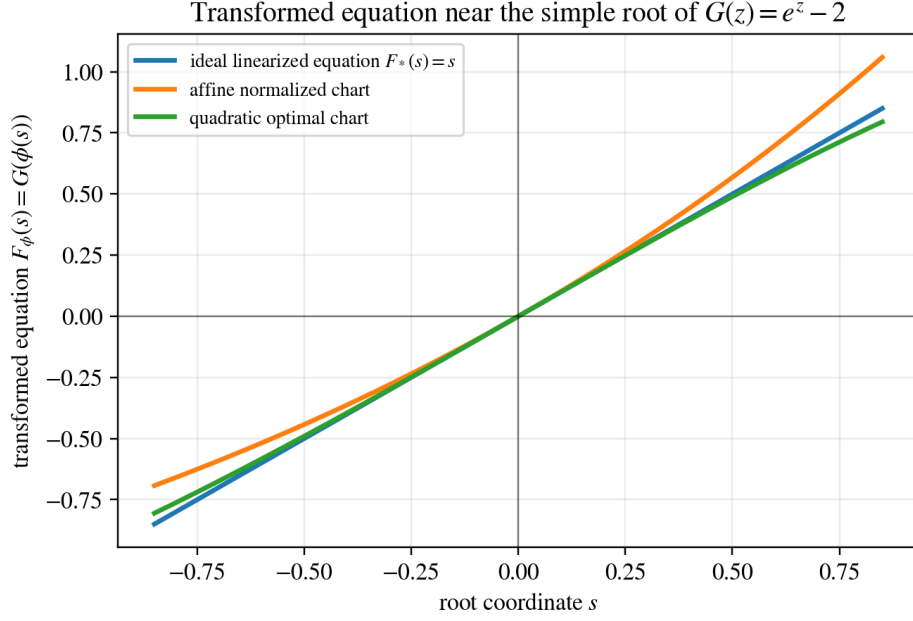


Figure 1: Transformed equations for the affine normalized chart, the curvature-balanced quadratic chart, and the ideal linearized model. The optimal chart is visibly flatter at the root.

For the quadratic family

$$\phi_\eta(s) = \rho + \frac{s}{2} + \eta s^2,$$

Theorem 4.3 gives

$$c_\eta = \frac{1}{4} + 2\eta,$$

so the optimal parameter is $\eta_* = -1/8$.

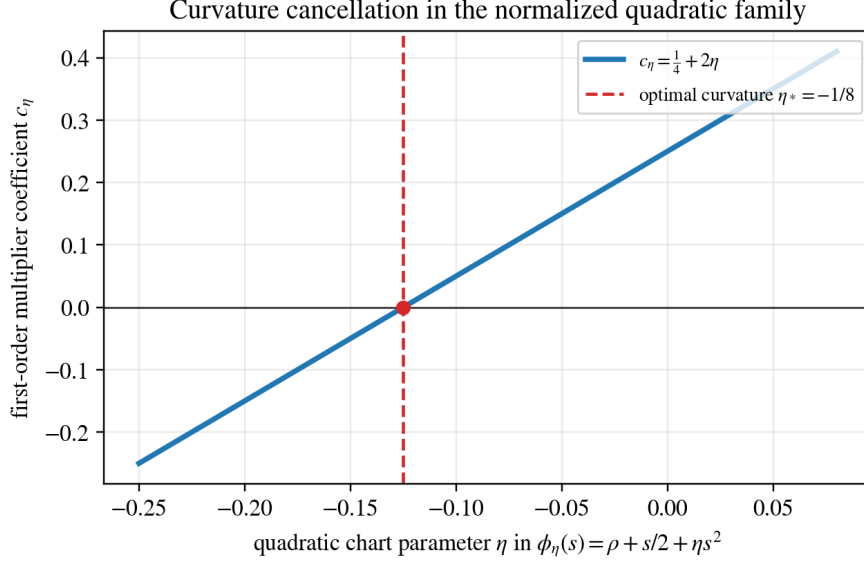


Figure 2: The first-order multiplier coefficient c_η in the quadratic chart family. The zero crossing at $\eta = -1/8$ is the curvature-balancing law.

Next, we compare the exact multiplier $\lambda_\phi(h) = P'_{\phi,a}(0)$ with anchor distance $h = |a|$. For the affine chart, $|\lambda_\phi(h)| \asymp h$. For the curvature-balanced chart, $|\lambda_\phi(h)| \asymp h^2$, exactly as predicted by Theorem 3.2.

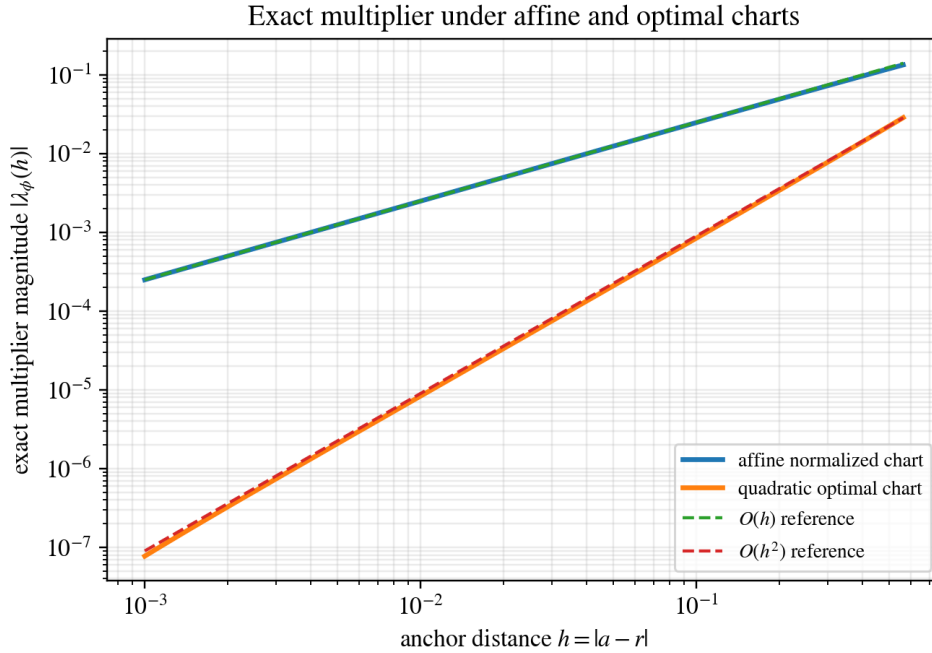


Figure 3: Exact multiplier magnitude versus anchor distance. The affine chart has linear behavior, while the optimal quadratic chart exhibits quadratic decay.

Finally, with anchor $a = 0.3$ and start $s_0 = 0.6$, the fixed-anchor orbits behave as follows.

chart	$ \lambda_\phi(0.3) $	iterations to $ s_n \leq 10^{-12}$	comment
affine normalized	7.3126×10^{-2}	11	linear regime
optimal quadratic	7.8149×10^{-3}	6	improved by curvature cancellation
exact inverse	0	1	ideal one-step model

Table 1: Fixed-anchor comparison for $G(z) = e^z - 2$, anchor $a = 0.3$, and start $s_0 = 0.6$.

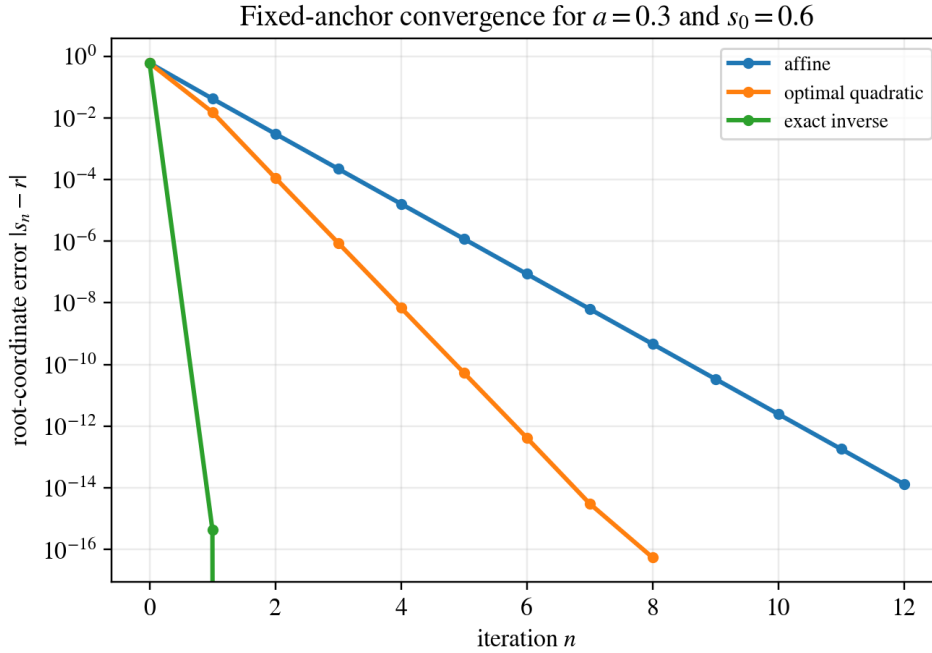


Figure 4: Root-coordinate errors for the same test problem. The curvature-balanced chart shortens the linear regime substantially, while the exact inverse chart collapses the problem in one step.

The point of these figures is not raw speed in the Newton-versus-Pandrosion sense. The point is that the chart alone changes the local geometry of the transformed equation, and the anchored iteration simply reflects that geometric change.

6 Discussion and outlook

The local inverse chart shows what “optimal” means in principle: the transformed equation should be as close as possible to the identity. The first-order optimality criterion then supplies the practical surrogate: cancel the quadratic defect of the transformed equation. This produces a clear geometric hierarchy.

- (1) **Affine scaling** changes the size of the multiplier coefficient.
- (2) **Curved chart optimization** can cancel the first nonlinear defect entirely.
- (3) **Exact inverse charts** give perfect local linearization when available.

This suggests several continuations.

- **Adaptive chart selection.** Instead of fixing one chart, one could update a local jet approximation of the inverse along the orbit.

- **Anchor–chart co-optimization.** The multiplier depends on both the anchor and the transformed equation. A natural next step is to optimize them jointly.
- **Several variables.** In higher dimension, the same viewpoint would replace scalar curvature cancellation by cancellation of the quadratic tensor in a root-normalized coordinate system.

The present note is therefore best read as a geometric preconditioning principle for anchored divided-difference iterations: before iterating, choose coordinates that flatten the transformed equation.

Acknowledgments

The figures in this manuscript were generated in Python with Matplotlib and incorporated into the final manuscript via L^AT_EX. The present note is intended as a continuation of the monomial and general anchored Pandrosion manuscripts.

References

- [1] I. Besevic, *A Monomial Pandrosion Scheme: Contraction, Chart Scaling, and Steffensen Acceleration for $z^p = x$* , Independent Mathematical Research, manuscript, May 2026.
- [2] I. Besevic, *Anchored Pandrosion for General Analytic Equations: Divided-Difference Fixed Points, Chart Selection, and Steffensen Acceleration*, Independent Mathematical Research, manuscript, May 2026.
- [3] A. C. Aitken, On Bernoulli’s numerical solution of algebraic equations, *Proc. Roy. Soc. Edinburgh* **46** (1926), 289–305.
- [4] J. F. Steffensen, Remarks on iteration, *Skandinavisk Aktuarietidskrift* **16** (1933), 64–72.
- [5] J. F. Traub, *Iterative Methods for the Solution of Equations*, Prentice–Hall, 1964.
- [6] A. S. Householder, *The Numerical Treatment of a Single Nonlinear Equation*, McGraw–Hill, 1970.
- [7] J. M. Ortega and W. C. Rheinboldt, *Iterative Solution of Nonlinear Equations in Several Variables*, Academic Press, 1970.
- [8] A. M. Ostrowski, *Solution of Equations and Systems of Equations*, 2nd ed., Academic Press, 1966.